

CLINICAL INTELLIGENCE TREND REPORT

RED: Provider Review Recommended

Patient ID: 934297-0122
Period: June 23 to June 30, 2024
Report Date: March 1, 2026
Resolution: Hourly aggregates (HR/RR min/avg/max)
Decision-support summary derived from longitudinal vital sign trends; interpret in clinical context.

SECTION 1 — Clinical Intelligence: Signal Summary

During the reporting period (June 23 to June 30, 2024), 127 hours of data were recorded. 7 episodic events were detected, including: Very Low Heart Rate, Low Heart Rate, Elevated Breathing Rate, High Heart Rate. Events occurred in clustered bursts rather than as isolated occurrences. Several episodes persisted for multiple consecutive hours (longest: 13 hours), suggesting sustained physiologic shifts rather than transient variation. Notably, periods of low heart rate coincided with elevated breathing rate. This concurrent pattern may reflect compensatory physiologic stress rather than isolated variation, and warrants closer clinical review.

Trend Assessment: Progressively unstable vital sign pattern
Overall Action Posture: Urgent provider review advised (per protocol)
Suggested Clinical Review Actions

- If sustained or recurrent low heart rate is observed (especially below 45 bpm for multiple hours): review symptoms, blood pressure, and heart rate-lowering medications; consider further cardiac evaluation if new or worsening.
- If low heart rate accompanies elevated breathing rate: assess for compensatory physiologic stress (low oxygen, fluid overload, infection); correlate with oxygen levels and clinical exam.
- If prolonged high heart rate (over 2 to 4 hours) or increasing daily frequency: evaluate for pain, infection, dehydration, anemia, or heart rhythm changes.
- If elevated breathing rate persists: correlate with oxygen saturation, work of breathing, and fluid status.
- If event frequency or duration is increasing over days: consider earlier provider notification, even without overt symptoms.

SECTION 2 — High Priority Episodic Events

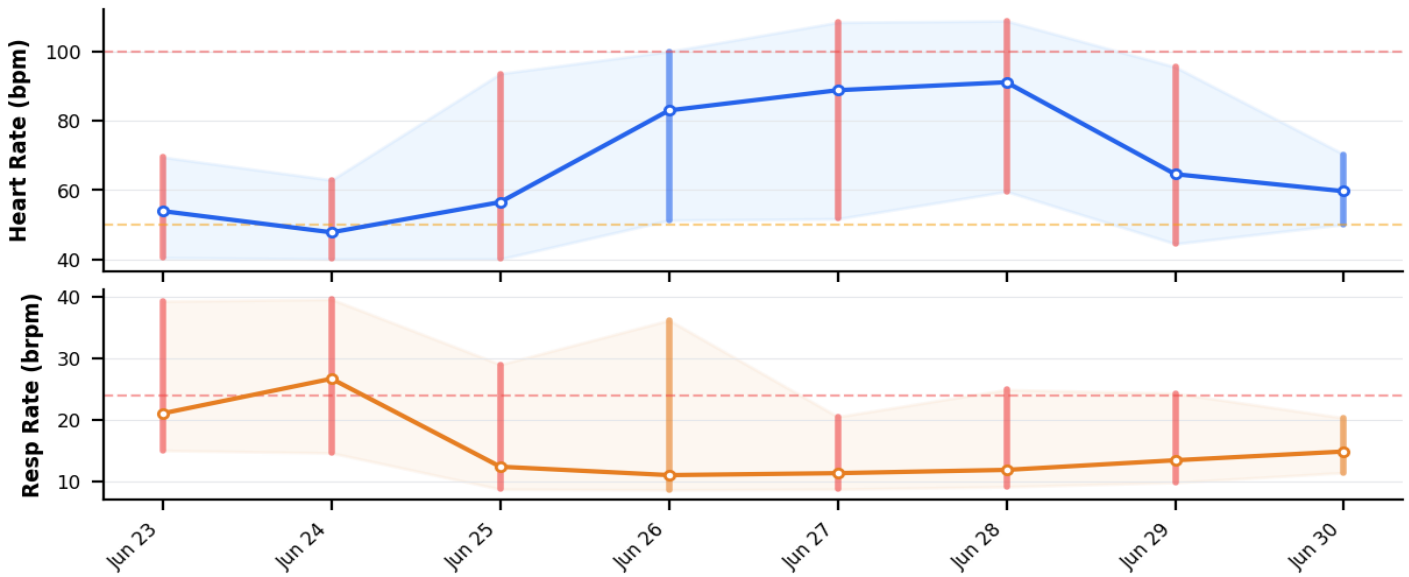
Event Type	Date / Time Window	Duration	Key Vitals	Comments
Very Low Heart Rate	06/23 21:00 to 06/24 10:00	13h	HR avg 48 Min HR 40 Max RR 40	Critical sustained pattern; urgent review advised (concurrent HR/RR)
Elevated Breathing Rate	06/24 00:00 to 06/24 11:00	11h	HR avg 47 Min HR 40 Max RR 40	Critical sustained pattern; urgent review advised (concurrent HR/RR)
High Heart Rate	06/27 20:00 to 06/28 02:00	6h	HR avg 102 Min HR 91 Max RR 19	Sustained deviation; review context
Very Low Heart Rate	06/25 01:00 to 06/25 05:00	4h	HR avg 45 Min HR 40 Max RR 29	Sustained deviation; review context
Low Heart Rate	06/24 11:00 to 06/24 11:00	1h	HR avg 48 Min HR 46 Max RR 35	Sustained deviation; review context (concurrent HR/RR)
Very Low Heart Rate	06/29 06:00 to 06/29 06:00	1h	HR avg 50 Min HR 44 Max RR 22	Sustained deviation; review context
Low Heart Rate	06/29 08:00 to 06/29 08:00	1h	HR avg 47 Min HR 47 Max RR 11	Brief deviation; continue monitoring

SECTION 3 — Heart Rate and Breathing Rate Statistics

Metric	Mean	Min	Max	P5	P95
Heart Rate Avg (bpm)	73.7	41.3	103.6	46.4	100.5
Heart Rate Min (bpm)	64.6	40.1	99.5	40.6	93.8
Heart Rate Max (bpm)	80.7	44.6	108.8	49.6	104.8
Breathing Rate Avg (brpm)	13.8	9.7	29.1	10.1	28.2
Breathing Rate Min (brpm)	10.7	8.6	18.4	8.7	16.9
Breathing Rate Max (brpm)	20.0	11.2	39.6	12.4	38.4

SECTION 4 — Visual Trend Analysis

Daily Vital Sign Ranges (Each bar shows daily min to max range; dots show daily average; red bars indicate days with flagged episodes; dashed lines show clinical thresholds)



Hourly Average Distribution (Shows how often each heart rate / breathing rate level was recorded; dashed lines show mean and threshold values)

